ABDUL MUQTADIR

Control Systems · Artificial Intelligence & Data · Embedded Systems & Robotics

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EDUCATION

National University of Sciences and Technology (NUST) Sep 2023 – Expected Jun 2027

Bachelor of Engineering in Avionics Engineering; CGPA: 3.74 / 4.00 (Semester 7) *Risalpur, Pakistan*

- **Honors & Coursework:** Dean's Honor List (Awarded across 5 semesters). Coursework in Control Systems Engineering, Digital Signal Processing (DSP), Microcontrollers & Embedded Systems, Artificial Intelligence Foundations, and Wave Propagation.

The Universal Millennium College Aug 2021 – May 2023

Cambridge International A-Levels — Mathematics (A), Physics (A), Chemistry (A) *Pakistan*

TECHNICAL SKILLS

Control Systems & Dynamics: Closed-Loop Digital PID Control, Linear Quadratic Regulator (LQR), State-Space Modeling, IMU 6-DOF Sensor Fusion (Kalman & Complementary Filters), Dynamic System Identification, MATLAB / Simulink.

AI, Data & Programming: Python (NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib), Machine Learning / Predictive Modeling, Data Analysis, C, C++, Embedded C, MATLAB, G-Code, Linux/Bash, Git.

Embedded Systems & Robotics: ESP32 (Dual-Core 240MHz), STM32, Arduino Uno, MPU6050 6-DOF IMU, BTS7960 43A High-Current H-Bridges, A4988 Stepper Drivers, NEMA Motors, Sensor Interfacing (I2C/SPI/UART).

Hardware & CAD Prototyping: CATIA V5 (3D Solid Modeling & Kinematics), AutoCAD, KiCAD/EasyEDA (PCB Layout), Digital Storage Oscilloscopes (DSOs), Soldering & Debugging.

RF, Microwave & Telemetry: CST Studio Suite 2025 (3D EM), Keysight ADS, S-Parameters (S11, S21), Vivaldi Antennas, ADALM-Pluto SDR, 915 MHz FM Telemetry.

SELECTED ENGINEERING PROJECTS

Two-Wheeled Inverted Pendulum Robot: LQR vs. Cascaded PID May 2026

MATLAB/Simulink, Embedded C++, MPU6050 6-DOF IMU, Control Theory *NUST CAE*

- Derived non-linear Lagrangian equations of motion and linearized state-space matrices around the unstable vertical equilibrium.
- Demonstrated physical degradation of theoretical state-feedback LQR under unobserved linear cart states; engineered and stabilized a physical prototype via cascaded PID firmware with anti-windup clamping and complementary sensor filtering.

Hardware-in-the-Loop (HIL) UAV Altitude Control & SDR Telemetry Jul 2026

Python Dynamics Plant, Closed-Loop PID Autopilot, ADALM-Pluto SDR *NUST CAE*

- Constructed a real-time HIL simulation coupling a simulated second-order UAV climb dynamics plant with a closed-loop digital PID autopilot.
- Modulated continuous altitude telemetry over a physical 915 MHz RF link using an ADALM-Pluto SDR; built receiver DSP pipeline with quadrature demodulation and single-pole IIR noise filtering to reconstruct the control feedback loop under noise.

Autonomous Heavy Competition Rover Apr 2025

ESP32 Dual-Core, Dual BTS7960 43A H-Bridges, 4S2P Li-Ion Battery Pack *Robotics R&D*

- Engineered heavy-duty robotic drivetrain utilizing dual BTS7960 43A H-bridges, high-torque JGA-25-370 geared motors, and custom 16.8V Li-Ion battery pack with real-time wireless telemetry.

Super Wideband (3–18 GHz) Conformal Vivaldi Antenna Aug 2026

CST Studio Suite 2025, Rogers RT6035HTC, CPW Transition *NUST CAE*

- Designed and simulated an ultra-compact planar and conformal Vivaldi antenna achieving a 6:1 bandwidth ratio with return loss S11 < -10 dB across 3.0 to 18.0 GHz; modeled aerodynamic conformal cylindrical bending matching UAV fuselage contours (isolation S21 < -25 dB).

EXPERIENCE & LEADERSHIP

Freelance Engineering Consultant Jul 2025 – Present

Independent Contractor — Embedded Systems, Controls & Firmware *Remote*

- Consulted on microcontroller firmware architecture, I2C/SPI sensor interfaces, and closed-loop control algorithms for international engineering clients.
- Conducted schematic reviews, circuit debugging, and component selection for high-reliability embedded hardware builds.

Head of Avionics Club Sep 2024 – Present

College of Aeronautical Engineering, NUST *Risalpur, Pakistan*

- Lead technical student projects, autonomous robotics workshops, and sensor telemetry testbeds for 40+ engineering students.

A-Level STEM Tutor (Mathematics & Physics) Nov 2022 – Present

Private Mentorship *Remote & In-Person*

- Mentored 20+ students in differential calculus, kinematics, wave mechanics, and electromagnetism, with students achieving top A/A* grades.